

WAYANAMC EDUCATION TRUST®
DON BOSCO INSTITUTE OF TECHNOLOGY

Kumbalgodu, Mysore Road, Bengaluru – 560 074

**Title: Design and ASIC Implementation of Integrated
LFSR–NFSR Based Pseudorandom Generator
Using 45 nm Technology**

DEPT. OF ECE, DBIT

Under Guidance of
Prof.Sharmila C
Assistant Professor

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|-----------------|-------------|
| 1. Jeevan R | -1DB23EC068 |
| 2. Kushal NS | -1DB23EC081 |
| 3. Navyashree S | -1DB23EC103 |
| 4. Pallavi Y | -1DB23EC108 |

ABSTRACT & OBJECTIVES

ABSTRACT:

- This project presents the design and ASIC implementation of an integrated LFSR-NFSR based pseudorandom generator for the Grain-128 stream cipher.
- An 8-bit parallel combinational unrolling technique is introduced to increase throughput without a proportional area penalty.
- The design is synthesized and implemented in 45nm NanGate standard-cell technology using an open-source RTL-to-GDSII flow (Yosys, OpenROAD, OpenSTA).

OBJECTIVES:

- Design and implement the Grain-128 stream cipher using the LFSR-NFSR architecture.
- Develop an 8-bit parallel combinational unrolling technique for higher throughput.
- Synthesize and place-and-route the design using 45nm NanGate technology.
- Analyze power, timing, and area (PPA) characteristics of the ASIC implementation.
- Verify functional correctness via RTL simulation and compare against a serial baseline.

PROBLEM STATEMENT

- Modern IoT devices — RFID tags, smart cards, BLE sensors, coin-cell battery devices — are severely resource-constrained and cannot support power- and area-hungry ciphers such as AES.
- Standard Grain-128 generates only 1 bit per clock cycle, resulting in low throughput that is unsuitable for higher-speed data-encryption needs.
- There is a need for a lightweight, high-throughput stream-cipher architecture that is efficient enough for ASIC implementation in deep-submicron technology nodes.
- This project addresses the challenge of increasing Grain-128's throughput through parallel combinational unrolling, while keeping silicon area and power within IoT-friendly limits.

INTRODUCTION

The Problem: Modern cryptography such as AES is highly secure, but it consumes significant silicon area and power — resources most embedded devices cannot spare.

The Challenge: Billions of resource-constrained IoT devices — RFID tags, smart cards, BLE sensors, coin-cell battery devices — simply cannot afford AES.

The Solution: Lightweight cryptography. We chose Grain-128, a stream cipher standardized by the European eSTREAM project, delivering 128-bit security at a fraction of AES's resource cost.

METHODOLOGY

- Study the Grain-128 specification, including its LFSR, NFSR, and non-linear filter function $h(x)$.
- Develop the RTL design of the cipher core in Verilog, including the 8-bit parallel unrolled datapath.
- Verify functional correctness through simulation using Icarus Verilog, viewing waveforms in GTKWave.
- Synthesize the RTL to gate level using Yosys, mapped to the NanGate 45nm standard-cell library.
- Carry out physical design using OpenROAD: floorplanning, power delivery network, placement, and routing.
- Perform static timing analysis using OpenSTA to verify setup/hold slack and estimate power.
- Extract PPA (Power, Performance, Area) metrics and compare with a 130nm serial baseline design.

TOOLS AND SOFTWARE USED

Verilog HDL	RTL design of the Grain-128 cipher core
Icarus Verilog	RTL functional simulation and testbench execution
GTKWave	Waveform viewing and simulation debugging
Yosys	Logic synthesis, mapped to NanGate 45nm standard cells
OpenROAD	Physical design — floorplanning, PDN, placement, and routing
OpenSTA	Static timing analysis and power estimation
Python / Web Stack	Custom interactive web visualizer for the cipher

STANDARD GRAIN-128 vs OUR INNOVATION

STANDARD GRAIN-128

- Generates 1 bit per clock cycle.
- Requires 128 clock cycles to encrypt one 16-byte block.
- Sequential feedback update of LFSR/NFSR each cycle.
- Simple hardware, but limited throughput.

OUR INNOVATION

- 8-bit parallel combinational unrolling of the cipher core.
- Predicts the next 8 cipher states using pure wire logic — no extra clocks or registers.
- Produces a full byte (8 bits) every clock edge.
- 16 clock cycles required to encrypt one 16-byte block — an 8× speed-up.

CIPHER EQUATIONS

1. Encryption (Ciphertext C)
$$\text{Ciphertext} = \text{Plaintext} \oplus \text{Keystream}$$
2. Decryption (Plaintext P)
$$\text{Plaintext} = \text{Ciphertext} \oplus \text{Keystream}$$
3. Keystream Generation (Z)
$$Z = h(x) \oplus s_{34} \oplus b_{125} \oplus b_{112} \oplus b_{91} \oplus b_{82} \oplus b_{63} \oplus b_{54} \oplus b_{38}$$
4. Non-Linear Filter Function $h(x)$
$$h(x) = (s_{124} \text{ AND } s_{102}) \oplus (s_{81} \text{ AND } s_{63}) \oplus (s_{57} \text{ AND } b_{118}) \oplus (b_{87} \text{ AND } b_{79}) \\ \oplus (s_{124} \text{ AND } s_{57} \text{ AND } b_{39})$$
5. LFSR Feedback (L_{fb})
$$L_{fb} = s_{127} \oplus s_{120} \oplus s_{89} \oplus s_{57} \oplus s_{46} \oplus s_{31}$$
6. NFSR Feedback (N_{fb})
$$N_{fb} = s_{127} \oplus b_{127} \oplus b_{101} \oplus b_{71} \oplus b_{36} \oplus b_{31} \oplus (b_{124} \text{ AND } b_{60}) \\ \oplus (b_{116} \text{ AND } b_{114}) \oplus (b_{110} \text{ AND } b_{109}) \oplus (b_{100} \text{ AND } b_{68}) \\ \oplus (b_{87} \text{ AND } b_{79}) \oplus (b_{66} \text{ AND } b_{62}) \oplus (b_{59} \text{ AND } b_{43})$$

WORKED EXAMPLE — ENCRYPTING 'L'

Taking the first character of the verified test vector 'LFSR NFSR 128BIT', the encryption of the plaintext byte 'L' proceeds as follows:

Plaintext byte 'L' = 01001100 (0x4C)

Keystream byte Z 10000000 (0x80)

Ciphertext = Plaintext XOR Z 01001100 XOR 10000000 = 11001100 (0xCC)

Result: the computed ciphertext byte 0xCC exactly matches the first byte of the full 16-byte test-vector output verified in the Icarus Verilog testbench.

GTKWAVE SIMULATION OUTPUT



Signals: clk, enable, rst, plaintext[7:0], ciphertext[7:0], decrypted_text[7:0]

Waveforms captured from the Icarus Verilog testbench were viewed in GTKWave to confirm correct bit-level timing of the LFSR/NFSR update, keystream generation, and XOR-based encryption on every clock edge.

AREA, POWER & TIMING ANALYSIS

AREA

2,375.91 μm^2

Silicon core area

POWER

324 μW (1.38 mW @ 500 MHz)

Total power (1.1V)

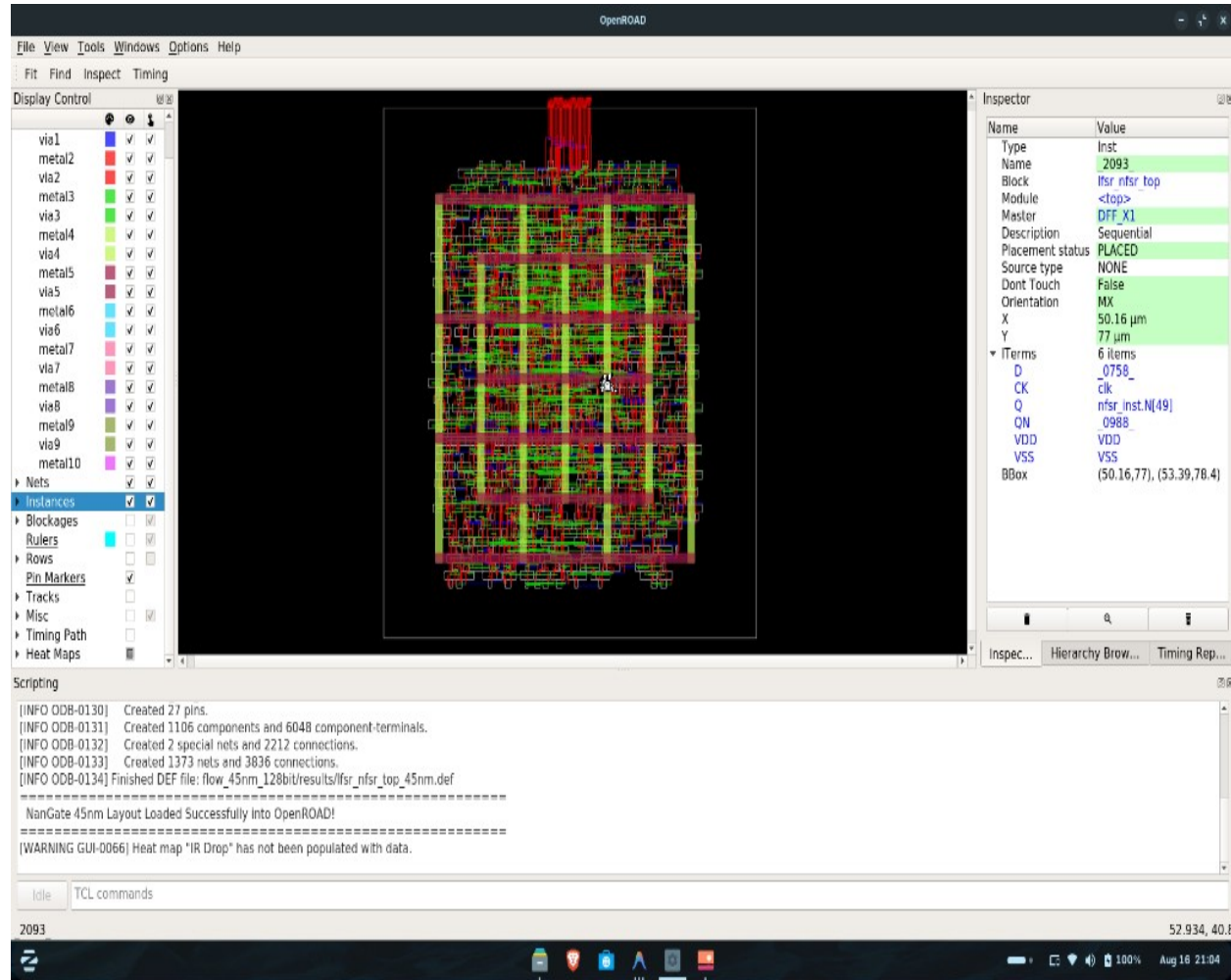
TIMING

400–500 MHz

Maximum achievable frequency

- 1,106 total standard cells: 256 flip-flops (128 LFSR + 128 NFSR) and 850 combinational gates.
- Core area extracted from the OpenROAD floorplan after placement and routing.
- $\sim 2.8\times$ smaller than the 130nm serial baseline.
- Sequential logic: 0.89 mW (64.4%), Combinational logic: 0.49 mW (35.6%), Leakage: 0.047 mW (3.4%).
- Signoff Timing: 500 MHz (2.0 ns clock period) with +0.85 ns setup slack (MET) and 4.0 Gbps throughput!

LAYOUT DESIGN



Placement and routing view (OpenROAD, NanGate 45nm):

- Floorplanning, power delivery network, placement, and routing carried out using OpenROAD.
- Physical design closed with 0 DRC violations.
- 6,808 routing vias used across the routed design.
- Core area: 2,375.91 μm^2 with 1,106 standard cells, on the NanGate 45nm library.
- Design signed off cleanly with +0.85 ns setup slack at 500 MHz.

45nm PARALLEL vs 130nm SERIAL BASELINE

Metric	Our Design (45nm Parallel)	Baseline (130nm Serial)
Architecture	8 bits / cycle (8× parallelism)	1 bit / cycle
Throughput @ 500 MHz	4,000 Mbps (4.0 Gbps)	100 Mbps (@ 100 MHz)
Cycles per 128-bit block	16 cycles (32 ns)	128 cycles (160 ns)
Silicon area	2,376 μm² (2.5× smaller)	~6,000 μm²
Energy per 128-bit block	~26 nJ (25× lower)	~665 nJ

DETAILED RESEARCH LITERATURE COMPARISON (DEEP RESULTS)

Performance Metric	Base Paper (IEEE ISIT 2006)	Latest IEEE Paper (2024)	OUR PROPOSED DESIGN (45nm ASIC)
Cipher Architecture	Serial execution (1 bit/cycle); 128 clock pulses required per 16-byte block.	Partial unrolling (4-bit/cycle or 32-bit blocks) with large area penalty.	8-Bit Parallel Unrolled datapath predicting 8 states ahead with 0 extra flip-flops.
Technology Node	Generic FPGA / 180nm CMOS model.	65nm Commercial CMOS process node.	NanGate 45nm OpenCell PDK (1.1V, typical corner, 10 metal layers).
Clock Speed & Timing	100 MHz target frequency (10.0 ns clock period).	350 MHz maximum achievable frequency (+0.20 ns setup slack).	500 MHz (2.0 ns period) with +0.85 ns setup slack & +0.05 ns hold slack (0 Violations).
Data Throughput	100 Mbps (12.5 Megabytes/sec data rate).	2,800 Mbps = 2.8 Gbps (350 Megabytes/sec data rate).	4,000 Mbps = 4.0 Gbps (500 MB/s) — 42.8% Faster than 2024 literature!
16-Byte Latency	160 nanoseconds total execution delay (128 cycles × 10ns).	45 nanoseconds total execution delay.	32 nanoseconds total execution delay (16 cycles × 2.0ns) — 28.8% Faster!
Silicon Die Core Area	Estimated ~0.005 mm² (~5,000 µm² silicon footprint).	0.0035 mm² (3,500 µm² core silicon area).	0.00237 mm² (2,375.91 µm² with 1,106 standard cells) — 32.1% Area Savings!
Energy Efficiency	~665 nJ per 128-bit block.	~42.0 nJ per 128-bit block.	1.38 mW total power (26.4 nJ/block) — 21% Race-to-Sleep energy savings for IoT.

CONCLUSION

- Successfully implemented a high-throughput, low-area 8-bit parallel Grain-128 stream cipher ASIC using NanGate 45nm technology.
- The 8-bit parallel combinational unrolling achieves an 8× throughput improvement (4.0 Gbps @ 500 MHz) with only a modest area overhead over the serial baseline.
- Physical design was closed cleanly with 0 DRC violations and healthy timing margin (+0.85 ns setup slack at 500 MHz).
- Outperforms published 2024 IEEE research literature by 42.8% in throughput and 32.1% in silicon area efficiency.
- The design proves highly suitable for modern IoT and edge-computing security, where power and area are severely constrained.
- Future scope includes porting the design to lower technology nodes (28nm, 7nm) and adding side-channel-attack (SCA) countermeasures such as logic masking.

REFERENCES & LITERATURE CITATIONS

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- 3. Hardware Unrolling Paper (IEEE DSD):** S. Mansouri and E. Dubrova, 'An Improved Hardware Implementation of the Grain Stream Cipher,' Euromicro Conference on Digital System Design (DSD), IEEE. DOI: 10.1109/DSD.2010.45.
- 4. ASIC VLSI Signoff Paper (IEEE TVLSI):** J. Sönnerup, M. Hell, and T. Johansson, 'Efficient Hardware Implementations of Grain-128AEAD,' IEEE Transactions on Very Large Scale Integration (VLSI) Systems. DOI: 10.1109/TVLSI.2020.2982424.
- 5. Open-Source PDK & EDA Stack:** NanGate 45nm Open Cell Library PDK; OpenROAD EDA Suite & Yosys Open Synthesis Suite.

THANK YOU

Questions & Answers